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10 / 10 submitted

Q1. Peak Element - List

Score: 10.00 TC: 1/1 passed

CODE

```
num=[1,3,20,4,1,0]
for i in range (1,len(num)-1):
    if num[i]>num[i-1]and num[i]>num[i+1]:
        print(num[i])
        break
```

OUTPUT

20

Q2. Second Largest and Second Smallest - Tuple

Score: 10.00 TC: 1/1 passed

CODE

```
a = [10, 20, 5, 40, 15, 30]
a.sort()
print(" %d"%a[1])
print(" %d"%a[-2])
```

INPUT

```
10
20
5
40
15
30
```

OUTPUT

```
10
30
```

Q3. Frequency of Each Element - List

Score: 10.00 TC: 1/1 passed

CODE

```
def frequency_of_elements(arr):
    freq = {}
    for x in arr:
        if x in freq:
            freq[x] += 1
        else:
            freq[x] = 1
    return freq
arr = [1,2,3,2,4,1,5,2]
freq = frequency_of_elements(arr)
keys = sorted(freq.keys())
for k in keys:
    print(k, "=", freq[k])
```

INPUT

1 2 3 2 4 1 5 2

OUTPUT

```
1 = 2
2 = 3
3 = 1
4 = 1
5 = 1
```

Q4. Missing Number - Tuple

Score: 10.00 TC: 1/1 passed

CODE

```
nums = (1,2,3,5,6)
n = len(nums) + 1

expected_sum = n * (n + 1) // 2
actual_sum = sum(nums)

missing = expected_sum - actual_sum
print(missing)
```

OUTPUT 4

Q5. Majority Element -List

Score: 10.00 TC: 1/1 passed

CODE

```
arr = [2,2,1,2,3,2,2]
n = len(arr)

maj = arr[0]
count = 0

for i in arr:
    occ = arr.count(i)
    if occ > n // 2:
        maj = i
        print(maj)
        break
```

OUTPUT 2

Q6. Sum & Average - Tuple

Score: 10.00 TC: 1/1 passed

CODE

```
t = (10,20,30,40,50)
s = sum(t)
avg = s / len(t) if t else 0
print(s)
print(avg)
```

OUTPUT 150
30.0**Q7. Element Exists -Tuple**

Score: 5.00 TC: 1/2 passed

CODE

```
a = (10,20,30,40,50)
x = 30

if x in a:
    print("Element found")
else:
    print("Element not found")
```

INPUT 30

OUTPUT Element found

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Q8. Maximum Element - Tuple

Score: 10.00 TC: 1/1 passed

CODE

```
num=(12,45,8,67,23)
largest=max(num)
print(largest)
```

INPUT

12 45 8 67 23

OUTPUT

67

Q9. Lucky Number

Score: 5.00 TC: 1/2 passed

CODE

```
def lucky(n):
    if n < 10:
        return n == 7
    s = 0
    while n > 0:
        s += n % 10
        n //= 10
    return lucky(s)

num = 34
if lucky(num):
    print("Lucky Number")
else:
    print("Not a Lucky Number")
```

INPUT

34

OUTPUT

Lucky Number

Q10. Remove Duplicate Elements - Tuple

Score: 10.00 TC: 1/1 passed

CODE

```
t=(10,20,30,20,40,10,50)
t=tuple(set(t))
print(t)
```

OUTPUT

(40, 10, 50, 20, 30)